Government of Tamilnadu

Department of Employment and Training

Course : TNPSC Group I Mains Material

Subject : Role and impact of Science and Technology in the development of India

Topic : Advancements in the fields of Robotics

© Copyright

The Department of Employment and Training has prepared the TNPSC Group-I Preliminary and Main Exam study material in the form of e-content for the benefit of Competitive Exam aspirants and it is being uploaded in this Virtual Learning Portal. This e-content study material is the sole property of the Department of Employment and Training. No one (either an individual or an institution) is allowed to make copy or reproduce the matter in any form. The trespassers will be prosecuted under the Indian Copyright Act.

It is a cost-free service provided to the job seekers who are preparing for the Competitive Exams.

Commissioner,

Department of Employment and Training.

ADVANCEMENT IN THE FIELD OF ROBOTICS

INTRODUCTION

Robotics is a branch of engineering that involves the conception, design, manufacture, and operation of robots. A robot is a machine designed to execute one or more tasks automatically with speed and precision. There are as many different types of robots as there are tasks for them to perform.

The word 'robotics' was used for the first time in print by Isaac Asimov. The term comes from a Czech word, robota, meaning 'forced labour.' The word robot first appeared in a 1920 play by Czech writer Karel Capek, Rossum's Universal Robots. In the play, the robots eventually overthrow their human creators. One early example of a robotic design dates back to about 1478: Leonardo the Vinci's car, a spring-driven autonomous system that was likely created to cause a sensation at court.

The first digitally operated and programmable robot, the Unimate, was installed in 1961 to lift hot pieces of metal from a die casting machine and stack them. Today, commercial and industrial robots are in widespread use performing jobs more cheaply and with greater accuracy and reliability than humans. They are also employed for jobs which are too dirty, dangerous or dull to be suitable for humans. Robots are widely used in manufacturing, assembly and packing transport, earth and space exploration, surgery weaponry, laboratory research, safety, and mass production of consumer and industrial goods.

Isaac Asimov's "Three Laws of Robotics"

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey orders given it by human being except where such orders would conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

EVOLUTION:

First-generation robots data from the 1970s and consist of stationary, nonprogrammable, electromechanical devices without sensors.

Second-generation robots were developed in the 1980s and can contain sensors and programmable controllers.

Third-generation robots were developed between approximately 1990 and the present. These machines can be stationary or mobile, autonomous or insect type, with sophisticated programming, speech recognition and synthesis, and other advanced features.

Fourth-generation robots ae in the research-and development phase and include features such as artificial intelligence, self-replications, self-assembly, and nanoscale size.

Some of the important components of Robots are as follows:

1. **Manipulator:** Just like the human arm, the robot consists of what is called a manipulator having several joints and links.
2. **End effector:** The base of the manipulator is fixed to the base support, and at its other free end, the end effector is attached. The End effector is expected to perform tasks normally performed by the palm and finger arrangements of the human arm.
3. **The Locomotion Device:** In the case of Human Beings the power for the movement of the arm, the palm and fingers are provided by muscles. For the robot, the power for the movement (locomotion) is provided by the motors. The motors used for providing locomotion in robots are of three types depending on the source of energy: Electric, Hydraulic or Pneumatic.
4. **The Controller:** The digital computer (both the hardware and the software) acts as a controller to the robot – the controller functions in a manner analogous to the human brain. With the help of this controller, the robot can carry out the assigned tasks. The controller directs and controls the movement of the Manipulator and the Effector. In other words, the controller controls the robot.

5. **The Sensors:** Without the data supplied by the sense organs, the brain would be incapable of intelligence. In other words, the controller (the computer) of the robot cannot do any meaningful task if the robot is not with a component analogous to the sense organs of the human body. Thus, the fifth and the most important component of the robot is the set of sensors. Sensors are nothing but measuring instruments which measure quantities such as position, velocity, force, torque, proximity, temperature, etc.

ADVANCEMENT

The field of robotics has greatly advanced with several new general technological achievements. One is the rise of big data, which offers more opportunity to building programming capability into robotic systems. Another is the use of new kinds of sensors and connected devices to monitor environmental aspects like temperature, air pressure, light, motion and more. All of this serves robotics and the generation of more complex and sophisticated robots for many uses, including manufacturing, health and safety, and human assistance.

The field of robotics also intersects with issues around artificial intelligence. Since robots are physically discreate units, they are perceived to have their own intelligence, albeit one limited by their programming and capabilities. This idea has generated new debates over traditional science fiction theories, such as Asimov's three laws of robotics, which address the interaction of humans with robots in some mechanized future.

POSITIVES:

In many instances, robots can perform work more efficiently than humans. They can work seven days a week, twenty-four hours a day, and thirty days a month without becoming bored or fatigued. The quality of their work can be checked and corrected immediately if found to be defective. Operating costs are low, and downtime is minimal. Thousands of people will be needed in the future to design, repair, and install robots. New jobs will be created, and new training programs will have to be developed to improve the use of robots.

Recent developments in the robotics world have made robots more user-friendly, intelligent, and most importantly affordable. With these benefits of robotics, it is no wonder that they have found jobs in every field. That is right, from industrial manufacturing to the medical field robots are being used.

The benefits of robots have increased their flexibility in being capable of performing a variety of tasks and applications. They are more precise and consistent than human workers. Robots also allow for increased production and profit margin because they can complete tasks faster.

Other benefits of robotics are that they can work in any environment, adding to their flexibility. Robots eliminate dangerous jobs for humans because they are capable of working in hazardous environments. They can handle lifting heavy loads, toxic substances, and repetitive tasks. This has helped companies to prevent many accidents, also saving time and money.

In the medical field, robots are used for intricate surgeries such as prostate cancer surgery. Robots can reach and fit where human hands cannot, allowing greater accuracy. Some robotic benefits in the medical field are less invasive procedures and less pain for the patient when recovering.

NEGATIVES:

Robots can replace humans in the labour force. They require a higher level of maintenance than do most existing jobs. Therefore, they require retraining or replacement of the humans now employed in that job. The initial cost of robots is excessive for small firms. The technology is relatively untested at this time, and downtime is expensive.

The robots cost much money in the maintenance and repair, The programs need to be updated to suit the changing requirements, the machines need to be made smarter, in case of the breakdown, the cost of repair may be very high, the procedures to restore lost code or data may be time-consuming and costly.

ROBOTICS IN INDIA:

In India, robotics has already reached an advanced stage where crucial sectors, such as healthcare, are utilizing the technology to enhance their offerings. Robotic medical assistants are being used to constantly monitor and interpret the vital stats of patients. These revolutionary medical assistants are also capable of analyzing patient symptoms and designing a source of treatment. From dispensing drugs to conducting screening tests, robots can now share a lot of responsibilities with humans.

Robots are also replacing human in warehouses where jobs are labour intensive and potentially dangerous. From heavy lifting to inventory collection, robots are capable of taking care of all warehousing tasks. In fact, thanks to the massive potential of robotics in warehousing, the Indian market for warehouse automation is projected to grow at a Compound Annual Growth Rate (CAGR) of 10-12% during 2015-2020 and is expected to reach $3.49 billion by 2020.

Furthermore, robots are also taking over mental tasks such as brooming and mopping floors through the advent of domestic robots. Domestic robots are incredibly intuitive robotic vacuum cleaners that clean your home at the push of a button. These futuristic robots are connected devices which can be accessed through mobile phones and AI-powered home assistants such as Google Home and Amazon Alexa. These powerful robot-vacuums can remove the smallest dust particles as well as large debris from homes. Without the slightest need for human intervention.

The best part about these robotic vacuum cleaners is that they can navigate through the house, avoiding stairs, cleaning under furniture and otherwise hard-to-reach spots. With advanced capabilities such as automatic dirt detection and smart navigation, these robots can significantly minimize the participation of humans in household chores.

Achievements:

Centre for Artificial Intelligence and Robotics (CAIR), Bangalore has developed a variety of controllers and manipulators for Gantry, SCARA and other types of robots. These were supplied to Public Sector units such as HAL and sister DRDO labs. CAIR

has gone on the develop a prototype Unmanned Ground Vehicle (UGV) with the aim of attaining autonomous capability. This involved in house construction of mobile robot platforms, integration of infrared sensors with the vehicle, and the development and integration of path planning software.

A useful offshoot of this work was the development of an intelligent wheelchair that would help physically challenged people both in hospitals and homes. One version of the wheelchair could be operated using human voice commands. Another was equipped with a camera system to get information about the surrounding space for its path planning. CAIR has developed CHATUR Robot with vision sensors which can pick objects in its visual field. Another premier institute where Robotics Research is going on is: Centre for Robotics and Mechanotronics, IIT, Kanpur.

Challenges:

There are many challenges that India must overcome before it is considered as a go to destination for robotics. One of the biggest challenges is cost and procurement of the required hardware and other electronic components to build a robot. Due to the extensive paperwork involved in importing hardware components into the country, not many commercial applications are ready to enter the market. Additionally, acquiring and retaining quality talent is one of the biggest challenges, as robotics is a multidisciplinary field. Another major challenge is that SME customers expect visible ROI before proactively adopting robotics.

The future of robotics in India:

The rapid transformation and advancement in the robotics sector in India come are refreshing development in the technological landscape. The perfect amalgam of A, Machine Learning and loT, robotics will undeniably take on scores of jobs currently being tackled by humans. The advancement in robotics is great news for humans as human workers can, therefore, focus on more creative, skill-based roles that will contribute to their pool of knowledge and fast-track their career growth.

As far as the future of robotics in India is concerned, the picture looks bright and replete with potential. Tremendous research is going into the sector, with technologists working towards inventions such as robotic pets, robotic security guards and robotic

palliative caretakers. Interestingly, Certain Al and machine-learning algorithms are even being taught to write poetry, create paintings and even dream.

ROBOTICS AND AUTOMATION:

Robotic technologies are extensively used across a range of sectors such as atomic energy, space metals, textiles, automotive, and manufacturing industries. Besides, the health sector in India has also initiated the use of robotic technology widely in operation theatres and even in rehabilitation centres to augment the quality of life. Robotics has proved to be a growing field and many avenues have opened in recent past. Many experts believe that robotics is best suited for industrial automation which includes manufacturing, packaging, and assembling. In fact, robotics and automation have the potential to revolutionize the industrial scenario by promising to bring the same result as computer systems have brought in services and other sectors. Robotics in the automation sector has proved to improve productivity, safety as well as the quality of the end product while allowing human operators to take up more value-added roles.

Robots are required everywhere to improve productivity. They are also being used in operation theatres and rehabilitation centres to augment the quality of life. Developed countries like Japan and American have been using robots to clean rooms, entertain etc. It is an ever growing field, and many avenues have opened up in recent past. Therefore, students who have the required skill sets will be favoured by such industries.

Robotics and Automation have the potential to revolutionize the industrial scenario. It promises to bring the same result as computer systems have brought in services and other sectors. However, many developing nations like India, Thailand, Indonesia, the Philippines are still to adopt robotics and automation in big way.

ROBOTICS IN OTHER COUNTRIES:

The relentless rise of robotics in manufacturing around the world that is expected for the next decade will affect the development of emerging market economies in different ways. The ‘East-Asian Tigers”, including China, are embracing robotics full-heatedly as the way to preserve the manufacturing intensity of their economies and build global

market share in medium and high-value-added industries, like automobiles, electronics, semiconductors and batteries. Germany and Japan are also manufacturing powers determined to maintain their manufacturing base through innovative engineering.

Most middle-income countries throughout South-east Asia and Latin America are lagging seriously behind in the robotics race and many find it increasingly difficult to move up the value chain to compete for market share in many highly-automated industries. These middle-income countries also face rising competition in low-value-added manufacturing from the new wave of industrializing countries with significantly lower labour costs, such as Vietnam, Bangladesh, India, Pakistan and Indonesia.

Until today, robot deployment has been highly concentrated in a few industries, namely motor vehicles electronics and metal-working. China has made robotics a key part of its government supported "Made in China 2025" plan, both regarding usage and domestic supply. The goal is to make China one of the world's top 10 most intensively automated nations by 2020, with 150 industrial robots per 10,000 employees, which compares to today's leader South Korea, will 531 robot units, the USA with 176 robot units and Germany with 301 robot units.

The deployment of manufacturing robots around the world is extreme in its geographic concentration. As the data from the International Federation of Robotics shows, the great majority of robots are being deployed in East Asia, North America and Germany. Middle-income countries with significant manufacturing bases, such as Brazil, Thailand, Malaysia, Turkey and South Africa, and European powers like France and the UK are being left far behind. The positive exceptions are Italy and Eastern Europe, perhaps because of the latter's close integration with German supply chain.

HUMANOID ROBOTS

Humanoid robotics is an important branch of biomimetic robotics and is not only associated with science and engineering disciplines but also deeply connected to social, legal, and ethical domains. A Humanoid may be defined as something that resembles or looks like a human and having characteristics like opposable thumb, ability to walk in an upright position, etc.

One important contribution to humanoid robotics was the zero-moment point (ZMP) stability theory introduced by Miomir Vukobratovic in the 1960s. The first humanoid statistically and later dynamically balanced robot, WABOT, by Ichiro Kato of Waseda University, Japan, was developed around the same time. Lately, Sophia, a social humanoid robot developed by Hong Kong – based company Hanson Robotics besides the first humanoid robot to be granted citizenship of any country. She has been prominently featured among speakers on the web page of the NASSCOM's WCIT - Nassscom India Leadership Forum 2018.

Features of Humanoid Robots:

The characteristics of Humanoid Robots include:

1. Self-maintenance
2. Autonomous learning
3. Avoiding harmful situations to people, property, and itself
4. Safe interacting with human beings and the environment

Purpose of Humanoid Robots

Humanoid robots are used a research tool in several scientific areas. Researchers need to understand the human body structure and behavior (biomechanics) to build and study humanoid robots. On the other side, the attempt to simulate the human body leads to a better understanding of it. Human cognition is a field of study which is focused on how humans learn from sensory information to acquire perceptual and motor skills. This knowledge is used to develop computational models of human behavior, and it has been improving over time.

Although the initial aim of humanoid research was to build better orthosis and prosthesis for human beings, knowledge has been transferred between both disciplines. A few examples are powered leg prosthesis for neuromuscularly impairs, ankle-foot orthosis, biological realistic leg prosthesis and forearm prosthesis. Besides the research, humanoid robots are being developed to perform human tasks like personal assistance, where they should be able to assist the sick and elderly, and dirty or dangerous jobs. Regular jobs like being a receptionist or a worker of an automotive

manufacturing line are also suitable for humanoids. In essence, since they can use tools and operate equipment and vehicles designed for the human form, humanoids could theoretically perform any task a human being can. Humanoid robots, especially with artificial intelligence algorithms, could be useful for future dangerous and distant space exploration missions, without needing to turn back around again and return to Earth once the mission is completed.

Future

Shortly we will require small, light and cheap robots that exhibit complex behaviours. For example, M2-M4 Macaco project has developed a portable system which is capable of emulating the head of different creatures both aesthetically and functionally. The pervasive robotics will concentrate its research on certain critical aspects like Object Analysis, Social Mechanisms and Navigation. Though the technology has advanced much in the field of Humanoid Robotics, there are still several problems which need attention. The technological brilliance of the humanoids is required to be sharpened more, and the shortcomings in the results must be dealt with properly. The field of Humanoid Robotics is a very promising one, and there are a lot of changes needed to be brought in this technology.

FIELDS OF APPLICATION

Robotics is an area of interest to human beings for more than one hundred years. On the other hand our perception of robots is influenced by the media and the international film industry (Hollywood). You may ask what robotics is all about? In my views, a robot's distinctiveness transforms depending on the atmosphere it works in.

Some of these are as follows: -

1. **Outer Space** – Robotic arms that are under the control of a human being are employed to unload the docking cove of outer-space shuttles to launch satellites or to build a space station.
2. **The Intelligent Home –** Robotic systems can nowadays scrutinize home safety, ecological circumstances and energy consumption. Door and windows can be unlocked mechanically, and an electrical device such as lights and A/C

can be pre-programmed to turn on. This helps residents to enjoy appliances irrespective of their mobility.

3. **Exploration** – Robots can enter the environments that are injurious to human beings. AN illustration is observing the atmosphere within a volcano or investigating our deep marine life. NASA has utilized a robotic probe for environmental study, ever since the early '60s.
4. **Farms** – Programmed robots are used by harvesters to cut and collect crops. Robotic milk farms are existing permitting workers to nourish and milk their cattle distantly.
5. **The Car Industry** – Robotic arms are used; these arms can execute numerous tasks in the car manufacturing and assembling procedure. They carry cut jobs such as sorting, cutting, welding, lifting, painting and bending.
6. **Disaster Areas** – Observation robots built-in with superior sensing and imaging gears. This robot can work in dangerous environments like urban site spoiled by earthquakes by inspecting floors, walls, and roofs for structural reality.
7. **Entertainment** – Interactive robots that show behaviours and education capability. One such robot is owned by SONY which moves around freely, responds to all your commands, carries your luggage and even responds to your oral instructions.
8. **Industrial robots** – These robots bring into play in an industrialized manufacturing atmosphere. Typically, these are articulated arms particularly created for applications like material handling, painting, welding and others. If we evaluate merely by the application then this sort of robots can also consist of some automatically guided automobiles and other robots.
9. **Domestic or household robots** – Robots which are used at home. This sort of robots consists of numerous different gears for example robotic pool cleaners, robotic sweepers, robotic vacuum cleaners, robotics sewer cleaners and other robots that can perform different household tasks.
10. **Medical robots** – Robots employed in medicine and medicinal institutes. First and foremost, surgical treatment robots. Also, some robotic directed automobiles and perhaps lifting supporters.

11. **Service robots** – Robots that cannot be classed into any other types by practice. These could be various data collecting robots; robots prepared to exhibit technologies, robots employed for research, etc.
12. **Military Robots** - Robots brought into play in military and armed force. This sort of robots consists of bomb discarding robots, various shipping robots, exploration drones. Flying robot drones are brought into play for the close watch in present time's modern armed force. In the future robotic aeroplane and automobiles could be employed to transmit petroleum, bullets, bombs, etc. or clear minefields.
13. **Space robots** – Robots employed in space as split apart type. This type of robots would consist of the robots employed on Canadarm that was brought into play in space Shuttles, the International Space Station, together with Mars explorers and other robots employed in space exploration and other activities.

ROBOTS AND EMPLOYMENT:

India's labour force faces tough competition from robots. About 20-30 per cent of employers in India anticipate a decrease in headcount due to automation taking over low-skill, monotonous jobs. At Infosys, for example, some 11,000 workers have already lost their jobs to automation, and 3,000 Wipro employees, faced the same fate after the company deployed Holmes, its Al project. These instances leave no doubt that IT industry jobs will downsize, losing 6.4 lakh jobs by 2021, according to HIS Research estimates.

And IT is not the only sector in jeopardy **Robotic Process Automation (RPA)** has started replacing workers in financial services and helped reduce costs by more than 50 per cent, because of improved accuracy and efficiency of data-intensive, repetitive tasks. These virtual systems can work round the clock and produce a faster turnaround. Axis Bank, ICICI and HDFC are among those who have adopted the technology for traditional functions such as passbook updating, cash deposits, verification of know-your-customer details, salary uploads and even loan processing and sales of financial products. The Indian auto sector, too, has adopted automation rapidly. For instance, Maruti Suzuki's factory in Manesar, Haryana has 7,000 workers and 1,100 robots. In agriculture, drones and robots will maximise yield and reduce damage to crops. High

employment generating sectors such as manufacturing, textiles and food processing services, too, are under threat.

Technology is now challenging human supremacy in many spheres by executing tasks faster and with smaller error margins. For instance, some Al can identify cancer more accurately than trained pathologists and sniff out fraudulent banking activity in milliseconds. Warehouse robots employed by ecommerce majors save time and work with higher precision. In addition to an increase in layoffs, mass hiring, too, has slowed down in tech firms as operations shift to smaller teams that can handle more sophisticated needs while robots take over simpler functions.

Cobots:

Following the 'Make in India' initiative, the government has also launched 'Skill India' programme to increase the labour productivity and ensure the workforce is acquainted with the new and upcoming technological advancements in the IT and automation sector of the manufacturing industry. Automation aims at efficient mass production of goods at a higher quality, which eliminates manual errors but these machines need to be supervised by humans. Industrial automation, for the longest time, was looked upon as operation of heavy machinery that had to be locked in cages, away from the workforce. The new line of light and easy-to-use robots was first introduced by Universal Robots in 2008, known as collaborative robots or cobots. These cobots will be teaming up with the work force in the manufacturing industry to make it more efficient and to increase productivity.

Experts say that the Make in India initiative will be a bigger success with the help of cobots as India has great potential in manufacturing and rising to a global platform. The deployment of these cobots can be a successful venture to the Indian manufacturers as they can be programmed by operations acting tremendous flexibility to meet human problem-solving needs and also ensures a safer environment for the employees.

Cobots give manufactures the opportunity to grow in measured steps so they can maintain competitiveness in a global landscape without taking on an undesired undue

risk. They can essentially be termed as the 'third hand' that helps the workforce in carrying out monotonous tasks.

These cobots were in line with industrial 4.0 and had managed to fulfil the idea of humans and robots co-existing in a workplace. Thus, the inclusion of collaborative robots in the 'Make in India' initiative can be a positive step in the future development of the manufacturing industry, and we will see a growth in employment and productivity along with the rise in technology, brining Indian manufacturing sector on a globally competitive platform.

Questions:

1. Discuss in detail about the evolution of robotics and its advancements.
2. What are the achievements and challenges in the field of Robotics in India?
3. Give a detailed account on Humanoid Robots.